



How does artificial intelligence adoption shape employee performance? A novel exploration of mimetic artificial intelligence performance through a hybrid approach based on PLS-SEM and ANN[☆]

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ABSTRACT

Within the context of Industry 5.0 and digital transformation, the rapid advancement of information technology has rendered artificial intelligence ubiquitous, presenting significant challenges and opportunities for the workforce. This study employs a hybrid Partial Least Squares Structural Equation Modeling (PLS-SEM) and Artificial Neural Network (ANN) approach to investigate employee performance within artificial intelligence-adoption work environments. Utilizing a three-wave experience sampling methodology, we collected 308 valid questionnaires from employees in Chinese internet companies. Our findings demonstrate that artificial intelligence adoption at work positively associated with employees' problem-solving efficacy, which in turn influences adaptive performance. Furthermore, stronger artificial intelligence adoption at work is associated with an increased employees' mimetic artificial intelligence performance through employee learning from artificial intelligence. Task-oriented leadership amplifies the effects of artificial intelligence adoption at work on both problem-solving efficacy and adaptive performance. Conversely, knowledge-oriented leadership strengthens the relationship between artificial intelligence adoption at work and employee learning from artificial intelligence, thereby developing mimetic artificial intelligence performance. This research contributes by introducing and measuring the novel concept of "mimetic artificial intelligence performance," extending the application of social comparison, learning and influence theories. The study offers valuable theoretical insights and practical implications for understanding and optimizing employee performance in artificial intelligence-driven workplaces.

1. Introduction

The rapid development of information technology has promoted the use of artificial intelligence (AI) in organizations. And the advent of artificial intelligence has been instrumental in empowering employees and organizations to navigate the risks associated with Industry 5.0 and digital transformations (Abbas et al., 2026). Employees' perceptions of artificial intelligence in the workplace are nuanced, with some viewing it as a beneficial tool and a productive approach to personal growth while others expressing concerns about its potential implications (Liang et al., 2022; Yin et al., 2024b). These divergent perceptions are likely to elicit disparate responses from employees when confronted with the ramifications of artificial intelligence adoption at work. In this context

of a uncertain work environment, employees may be driven to engage in social comparisons to accurately assess their abilities (Festinger, 1954). Concurrently, the anthropomorphism theory in human-computer interaction (HCI) posits that the presence of anthropomorphic characteristics in artificial intelligence could result in employees perceiving it as a collaborator or colleague (Nass and Lee, 2001). This establishes a foundation for comparative analyses between employees and artificial intelligence (Wang et al., 2023). In light of the dual nature of artificial intelligence adoption at work, this study posits that when employees perceive artificial intelligence as a threat, they may engage in downward social comparisons to positively evaluate their abilities, thereby mitigating the occurrence of negative psychological effects. When perceiving opportunities, employees may engage in upward assimilation

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social comparisons with artificial intelligence, which is driven by the pursuit of direction and information for self-improvement. Furthermore, the manner in which individuals engage with social comparisons may result in divergent attitudes and behaviors concerning artificial intelligence. [Tschoopp et al. \(2023\)](#) pointed that individuals' attitudes toward artificial intelligence are diverse, individuals may regard artificial intelligence as an assistant, a transactional partner, or a friend. [Li et al. \(2024\)](#) initially introduced the concept of “employees learning from artificial intelligence”, in which employees regard artificial intelligence as a role model to enhance their own abilities. Drawing on previous studies, this paper, starting from the social comparison between employees and artificial intelligence as well as the dual effect of artificial intelligence adoption at work, unifies employees' two attitudes toward artificial intelligence, namely “utilizing artificial intelligence” and “learning from artificial intelligence”. We point out that when employees perceive threats, the downward comparisons with artificial intelligence will further prompt them to regard artificial intelligence as an assistant and utilize it to improve their adaptive performance; when employees perceive opportunities, the upward assimilation comparisons with artificial intelligence will further encourage them to learn from artificial intelligence to enhance their mimetic artificial intelligence performance.

On the one hand, when employees perceive artificial intelligence adoption at work as a threat, comparing themselves to artificial intelligence can help them develop a positive view of their abilities and motivate them to use artificial intelligence to enhance their adaptive performance. Adaptive performance refers to the behaviors employees exhibit in response to changes in the organizational environment. These changes are dynamic and emphasize employees' ability to adapt flexibly ([Shoss et al., 2012](#)). Adaptive performance requires employees to self-adjust to changes in the work environment brought about by artificial intelligence adoption at work and acquire resources to mitigate environmental change impacts ([Park and Park, 2021](#)). [Bandura \(2001\)](#) noted that higher self-efficacy promotes greater work engagement and investment when faced with highly variable and challenging work demands, enabling employees to perform more adaptively. Additionally, artificial intelligence's capabilities in collecting and providing information and processing data can help employees improve work efficiency. Thus, when employees view artificial intelligence as an assistant and utilize it, the supportive work assistance provided by artificial intelligence can enhance their confidence in problem solving, i.e., their problem-solving self-efficacy, thereby further improving their adaptive performance.

On the other hand, [Zhou et al. \(2024\)](#) pointed out that employees would use the knowledge and information platform provided by artificial intelligence to learn new skills and thereby improve their work performance. This shows that in the context of artificial intelligence adoption at work, when interacting with artificial intelligence, employees may not only utilize artificial intelligence but also generate learning motivation and behavior to improve their work performance. Previous studies have primarily emphasized the role of artificial intelligence as an assistant or service provider in the employee learning process, while neglecting the exemplary effect of artificial intelligence. Although [Li et al. \(2024\)](#) proposed the concept of employees learning from artificial intelligence, they did not conduct further research on the antecedents (e.g., human-computer social comparison) and consequences (e.g., higher work performance) of employees learning from artificial intelligence. To address this gap, this study integrates social learning theory ([Bandura, 1986](#)), suggesting that when employees perceive opportunities for artificial intelligence adoption at work, they engage in upward assimilation comparisons with artificial intelligence. Following such comparison, employees not only develop motivation to learn from artificial intelligence but also apply the skills they have learned in their daily work to enhance mimetic artificial intelligence performance, which refers to the artificial intelligence-like skills and behaviors employees exhibit after directly learning from and imitating

artificial intelligence, such as the logical thinking of chatbots and the operational actions of artificial intelligence robots ([Li et al., 2024](#)).

As illustrated, employees' psychological states and work behaviors demonstrate uncertainty in the context of the dual effect of artificial intelligence adoption at work. This study further posited that such uncertainty can be influenced by differences in leaders' goal orientations. In accordance with the principles of social influence theory ([Kelman, 1958](#)), the psychological states and behaviors of individuals frequently undergo alterations in response to the influence of others, contingent upon the significance and proximity of these individuals. Therefore, as posited by [Ford et al. \(2021\)](#), leaders' goal orientations, which they demonstrate through their interactions with employees, have the capacity to positively influence employee work performance. For instance, knowledge-oriented leaders emphasize knowledge sharing within the organization and encourage employees to develop themselves through learning. In the context of knowledge-oriented leadership, employees tend to perceive fewer threats and more opportunities from artificial intelligence ([González-Mohino et al., 2024](#)). The leader's endorsement of learning fosters employee engagement in upward assimilation comparisons with artificial intelligence, leading to the acquisition of skills that enhance mimetic artificial intelligence performance. In contrast, task-oriented leaders prioritize the efficient completion of organizational tasks ([Rüzgar, 2018](#)). In the context of task-oriented leadership, employees tend to perceive both stronger threats and weaker opportunities associated with artificial intelligence adoption. The pressure to efficiently complete tasks and task-related support from leaders (such as reasonable task allocation and guidance) prompt employees to actively compare themselves with artificial intelligence to gain psychological security, while at the same time utilizing artificial intelligence to improve their problem-solving efficacy and adaptive performance.

Based on the aforementioned gaps, this study applies social comparison and influence theory to explore when and how employees use and learn from artificial intelligence using survey data collected from internet companies in China. This study aims to make the following contributions. First, this study introduces a new concept (mimetic artificial intelligence performance) and developing a scale to measure it. Second, this study proposes a research model in which artificial intelligence serves as an assistant that helps employees improve their problem-solving efficacy and learn from artificial intelligence. Furthermore, the research framework is based on comparisons between employees and artificial intelligence, and points out the causes, processes, and outcomes of human-artificial intelligence social comparison. Consequently, the present study contributes to the expansion of the theoretical framework of social comparison and learning theory. Thirdly, the study considers leader orientations (e.g., task-oriented leadership) as moderating variables, exploring the different impact mechanisms of task-oriented and knowledge-oriented leaders on employee performance. This extends the boundary conditions of the impact of artificial intelligence adoption at work on employee performance. The study also provides theoretical and practical insights for future research on employee performance in artificial intelligence adoption at work scenarios.

The remainder of this article is organized as follows. Section 2 contains the theoretical analysis and research hypotheses, Section 3 contains the research design, Section 4 contains the empirical results, Section 5 contains the general discussions, and Section 6 contains the conclusions.

2. Theoretical background and hypothesis development

2.1. Employee-artificial intelligence interaction and its dual effects

As the Industry 5.0 continues to deepen and artificial intelligence technology continues to advance, artificial intelligence technologies centered on artificial intelligence entities or generative chatbots (such as Chatgpt) are becoming increasingly prevalent in organizations. The

boundaries between human-to-human interaction and human-artificial intelligence interaction (HAI) are beginning to blur (Chaturvedi et al., 2023). HAI refers to the process of information exchange between humans and bots integrated with artificial intelligence technology through language or text communication. The primary differences from traditional human-to-human interaction lie in the unique anthropomorphic attributes of the artificial intelligence interaction partner and the automated attributes of artificial intelligence responses, such as real-time connectivity (ubiquitous connectivity), personalization, and information transparency (Baabdullah et al., 2022). The Computers Are Social Actors (CASA) framework highlights that during HCI, due to computers' inherent anthropomorphic attributes (such as anthropomorphic appearance and emotional expression), individuals tend to perceive computer as human partners, actively attributing human characteristics to computers (Konya-Baumbach et al., 2023; Nass and Lee, 2001). In recent years, as artificial intelligence automation has advanced, its anthropomorphic attributes have become further enhanced (e.g., Deep Seek's anthropomorphic deep thinking and responses). Researches have shown that the anthropomorphic voice of chatbots may enhance employees' perception of the social presence and trust in chatbots, while the eye movements and facial expressions of physical robots may also make employees feel the robot's attention and emotional communication (Liu and Wei, 2021; Pitardi and Marriott, 2021). Additionally, Konstantoulaki et al. (2022) found that artificial intelligence chatbots' automated functions, such as usefulness, problem-solving, and customized responses, can have a substitution effect on human service staff and enhance customer experience and satisfaction. Therefore, artificial intelligence's human-like and automated functions narrow the gap between human-to-human interaction and HAI, enabling artificial intelligence to fully serve as an assistant or colleague to employees in work settings.

The dual effects of HAI refers to the phenomenon where the automation and human-like characteristics of artificial intelligence can lead employees to perceive negative threats such as job insecurity, occupational burnout, and emotional exhaustion, as well as positive opportunities such as enhanced self-efficacy, work efficiency, and creativity through the use of artificial intelligence (Liang et al., 2022; Liu and Cheng, 2025). Yin et al. (2024b) noted that when employees perceive artificial intelligence adoption as a threat to their work (STARA awareness), they may actively adopt avoidance behaviors (e.g., reducing innovative behaviors); conversely, when employees perceive the benefits and potential opportunities of artificial intelligence adoption for their work, they may actively utilize artificial intelligence to preserve and acquire cognitive resources to enhance their innovative self-efficacy. Therefore, employees possess a certain degree of perception and self-regulation capabilities when faced with artificial intelligence adoption at work, but the specific mechanisms of self-regulation warrant further exploration and research. In this regard, based on artificial intelligence anthropomorphism theory and the perspective that artificial intelligence can serve as a colleague, employees may regulate the dual effects of artificial intelligence colleagues through social comparisons with them (Wang et al., 2023).

2.2. Social comparison theory in HAI

Individuals' misperceptions and misjudgments of their own abilities may cause them harm to some extent (Festinger, 1954). Therefore, the primary purpose of individuals engaging in social comparison is to obtain an accurate and stable assessment of their own characteristics and abilities. Subsequent researches have shown that individuals may engage in social comparison for more diverse purposes, such as to gain self-enhancement through downward contrast comparison, maintaining positive evaluations of their own characteristics, or gaining direction and information for self-improvement through upward assimilation comparison (Crusius et al., 2022; Taylor and Lobel, 1989; Wills, 1981). According to social comparison theory, employees may tend to seek out

colleagues who are similar to themselves in key dimensions (such as ability and personality) as standards for social comparison, as similar comparison objects often help them make more accurate assessments; alternatively, to understand an unfamiliar dimension, employees may compare and learn from colleagues who are dissimilar in that dimension (Wood, 1989). When interacting with artificial intelligence, its anthropomorphic attributes confer a sense of similarity to employees, while its automated attributes—such as data processing and decision-making logic—highlight the distinct differences between artificial intelligence and humans (Jarrahi, 2018). Therefore, based on their recognition of similarities and differences between themselves and artificial intelligence colleagues in terms of capabilities and personality, employees may be motivated to engage in multidimensional social comparisons with them (Wood, 1989; Yin et al., 2024b).

This study suggests that artificial intelligence has the potential to serve as a colleague based on its anthropomorphic attributes, but also acknowledges that there are significant differences between artificial intelligence colleagues and human colleagues. And these differences may be the primary reason why employees actively choose artificial intelligence as a social comparison object to regulate the dual effect rather than choosing more intimate human colleagues. According to social comparison theory, social comparisons between individuals can primarily be categorized into upward assimilation comparisons and downward contrast comparisons. The choice of comparison method depends on the individual's purpose for conducting the social comparison (e.g., self-enhancement and self-improvement) (Crusius et al., 2022; Festinger, 1954). Correspondingly, different comparison methods yield different comparison outcomes, and the effects of these outcomes are relatively complex and multifaceted. Consider the hubristic pride employees feel after downward contrast comparisons with colleagues who perform worse in certain stable areas, such as talent. This sense of superiority, while helping individuals maintain a positive self-evaluation, may also prompt them to exhibit antisocial behavior, leading to negative evaluations from the compared colleagues and poorer interpersonal relationships (Tracy et al., 2009). Similarly, if employees engage in upward assimilation comparisons with colleagues who perform better, they may discover that the colleagues' superior performance is due to factors that are difficult to change, such as talent, job position, or work resources. This may further cause employees to experience negative emotions, such as malicious envy toward colleagues, and to deviate from the original purpose of seeking self-improvement information (Lange and Crusius, 2015). As can be seen, in actual interpersonal social comparison processes, the outcomes individuals obtain after engaging in social comparisons can be uncertain and may lead to unintended consequences. The underlying cause of this phenomenon is likely the emotional relationships between individuals. For example, Pinkus et al. (2008) found that after engaging in upward assimilation comparisons with intimate partners, individuals held more positive views of their partners and relationships compared to downward contrast comparison scenarios. In contrast to interpersonal relationships, the emotional connection between artificial intelligence and humans is much weaker. Although recent trends toward artificial intelligence anthropomorphism and advancements in algorithms have enabled artificial intelligence to play an increasingly significant role in people's daily lives, at its core, artificial intelligence remains a machine and tool created by humans. While artificial intelligence and humans share certain similarities at present, it remains challenging for them to develop intimate relationships akin to those between human partners (e.g., romantic relationships, Lockwood et al., 2004). This limitation restricts the depth of communication between artificial intelligence and humans but also prevents uncertain comparison outcomes arising from emotional relationships when humans compare themselves to artificial intelligence in a social context. Still taking hubristic pride as an example, employees may develop hubristic pride after comparing themselves with artificial intelligence based on their unique human advantages (such as empathy and creativity). This pride helps them maintain a positive image without

triggering malicious envy from artificial intelligence or deteriorating their relationship with it. Meanwhile, when employees engage in upward assimilation comparisons with artificial intelligence, even if they discover artificial intelligence's advantages (such as computational ability and logical reasoning), due to the algorithmic "transparency" of artificial intelligence as a machine, employees can identify areas for improvement without developing negative self-evaluations or malicious envy toward artificial intelligence. In other words, when engaging in social comparisons with artificial intelligence, employees are more likely to obtain a relatively certain comparison result, thereby achieving the original purpose of comparisons. Additionally, conducting social comparisons requires employees to obtain specific judgment-relevant information about themselves and the standard (Crusius et al., 2022). Compared to investing significant resources to understand human colleagues, employees can more conveniently and quickly obtain information about artificial intelligence through simple question-and-answer interactions with it. Based on the above discussions, this study proposes that employees are more likely to choose artificial intelligence colleagues as a social comparison standard rather than human colleagues.

Previous studies primarily focused on the negative effects of social comparison between employees and artificial intelligence or machines, such as the competitive pressure it imposes on employees (e.g., Wang et al., 2023). This research suggests that to mitigate the perceived threat of artificial intelligence colleagues to their own jobs, employees may engage in downward contrast comparisons with artificial intelligence colleagues, thereby viewing artificial intelligence colleagues as artificial intelligence assistants and leveraging them to boost self-efficacy; to seize the potential opportunities for self-improvement presented by artificial intelligence colleagues, employees may engage in upward assimilation comparisons with them to identify areas for improvement while viewing artificial intelligence colleagues as role models and learning from them to enhance work performance (Li et al., 2024). Thus, social comparisons between employees and artificial intelligence colleagues can help employees navigate the dual effects of artificial intelligence adoption at work and exert more positive influences on employee work performance. Building on this theoretical foundation, we can further explore the positive impacts of artificial intelligence adoption at work on employee work performance.

2.3. Artificial intelligence adoption at work and employee adaptive performance: The mediating role of employee problem-solving efficacy

When using artificial intelligence, the comparison type or dimension chosen by employees depends on their own comparison objectives: self-improvement and self-enhancement. This section primarily discusses the reactions and behaviors triggered when employees feel threatened by artificial intelligence adoption and engage in downward contrast comparisons with artificial intelligence for the purpose of self-enhancement.

To address perceived threats associated with artificial intelligence adoption at work, employees tend to engage in downward contrast comparisons with artificial intelligence to gain a sense of superiority in terms of capability. Therefore, during the process of capability comparison, employees may intentionally select dimensions where humans clearly have an advantage, such as emotional decision-making in uncertain situations, as the content of comparison. Once the comparison results demonstrate that artificial intelligence indeed falls short of humans in these areas, employees may experience more positive feelings and self-enhancement, thereby exhibiting greater confidence in their own capabilities and higher levels of self-efficacy (Klein, 2003). In such a scenario, with the guidance of leaders, employees can interact with artificial intelligence in a more positive mindset, and their frequency of artificial intelligence use and familiarity with artificial intelligence will also increase. Employees may gradually recognize that while artificial intelligence may be inferior to humans in decision-making and logical reasoning (Jarrahi, 2018), it can still enhance their confidence and efficacy in problem-solving by performing simple, repetitive tasks or

providing assistance to employees engaged in complex tasks, such as information processing and analysis (Haenlein and Kaplan, 2019). Therefore, this study posits that after comparing themselves with artificial intelligence, employees may develop a tendency to view artificial intelligence as an assistant due to the enhanced confidence in their own abilities. Furthermore, they may feel more confident addressing issues encountered in new task scenarios because they feel assisted by artificial intelligence, thereby enhancing their problem-solving efficacy.

Problem-solving efficacy, as a self-perception variable, reflects an individual's confidence in their own abilities when faced with complex and challenging problems (Bandura, 1997; Lin, 2023). It enhances an individual's positive beliefs in striving toward desired goals. In artificial intelligence adoption at work contexts, employees may seek to mitigate the negative impacts of artificial intelligence, such as uncertainty, by narrowing their scope of work and concentrating resources on addressing core work issues to enhance their adaptability (Bindl et al., 2019). Hence, when employees possess higher problem-solving self-efficacy, the increased confidence and efficiency in problem-solving enable them to better adapt to changing task environments. Adaptive performance refers to the adaptive behaviors employees exhibit in response to organizational environmental changes (Shoss et al., 2012). It represents the proficiency with which individuals adjust their behaviors to meet changing task demands, encompassing dimensions such as problem-solving, coping with uncertain or unpredictable work scenarios, and managing work-related stress (Pulakos et al., 2000). Thus, employees utilizing artificial intelligence to adjust their work behaviors and enhance their efficiency in addressing changing issues essentially embody adaptive behaviors. Furthermore, employees with higher problem-solving efficacy are often perceived as effective task performers within the organization when facing changes in work and task environments brought about by artificial intelligence, as they exhibit more positive and confident attitudes toward problem-solving (Pulakos et al., 2000). High-performing task executors typically exhibit more flexible adaptive behaviors in problem-solving, uncertainty, stress, or crisis management (Mo et al., 2024; Park and Park, 2019). Based on these, this study posits that higher problem-solving efficacy can lead employees to exhibit more task-related adaptive behaviors (Jundt et al., 2015), i.e., higher adaptive performance. And the following hypothesis is proposed:

H1. Artificial intelligence adoption at work is associated with an increased employees' adaptive performance by improving their problem-solving efficacy.

2.4. Artificial intelligence adoption at work and employees' mimetic artificial intelligence performance: The mediating role of employees learning from artificial intelligence

The dual effects of employees using artificial intelligence manifests itself in two ways: on the one hand, employees may feel pressure and threat due to artificial intelligence adoption at work; on the other hand, they may also recognize the opportunities for self-improvement that artificial intelligence brings. This section primarily discusses the reactions and behaviors triggered when employees perceive the opportunities brought by artificial intelligence and engage in upward assimilation comparisons with artificial intelligence for the purpose of self-improvement.

Drawing on social comparison theory, when employees perceive the opportunities brought by artificial intelligence adoption at work, they tend to view artificial intelligence as a tool for self-improvement. And employees may engage in upward social comparisons with artificial intelligence to identify dimensions where artificial intelligence outperforms them, thereby determining the direction and standards for their own progress. Social learning theory suggests that individuals learn through observing and imitating others or by setting examples (Bandura, 1986). In the social learning process, individuals can observe the specific behaviors of outstanding role models and their differing outcomes in

specific social environments and actively learn from them to make their own performance more similar to that of the role models. In traditional work settings, employees primarily learn from colleagues and leaders. Li et al. (2024) argued that due to artificial intelligence's anthropomorphism and its potential to serve as an employee colleague, employees may learn from artificial intelligence when they observe positive outcomes (e.g., rational decision-making logic, advanced algorithmic models) produced by artificial intelligence as a demonstrator. Thus, social learning theory and upward social comparison share similarities, both emphasizing the importance of observing and imitating superior objects to facilitate self-improvement. Additionally, artificial intelligence's advantages in information collection, knowledge generation, and accessibility can strengthen the flow of knowledge, support organizational learning (Abbas et al., 2026). This is crucial for employees to adjust workflows to address errors, obstacles and facilitates employee learning (Dorta-Afonso et al., 2024). Based on this, this study combines upward comparison with social learning theory to argue that employees, after engaging in upward comparison with artificial intelligence, will further view artificial intelligence as a learning model and seek to learn from it.

When learning from artificial intelligence, employees closely observe and remember the behaviors or skills demonstrated by artificial intelligence, and after learning, they attempt to apply these behaviors or skills in their work (Bargh et al., 1996; Rumjaun and Narod, 2020). Previous studies have shown that when employees face artificial intelligence-related stress or work in a digital environment, they utilize artificial intelligence (e.g., artificial intelligence-provided knowledge collection and sharing platforms) to acquire new knowledge and skills (e.g., chatbot logic, algorithms, and language) (Raisch and Krakowski, 2021). This study posits that, compared to learning through artificial intelligence, when employees learn directly from artificial intelligence, the new skills they observe, remember, and imitate will be closer to artificial intelligence itself and exhibit more pronounced artificial intelligence characteristics (e.g., the operational behavior of collaborative robots, the decision-making logic of chatbots) (Li et al., 2024). Additionally, researches indicated that employees' acquisition and mastery of artificial intelligence-related knowledge can promote the development of computer self-efficacy (CSE) or artificial intelligence self-efficacy (AISE) (Wang and Chuang, 2024; Yin et al., 2024b). This sense of self-efficacy encourages employees to invest more resources and actively apply the artificial intelligence-related skills they have learned to tackle more challenging work tasks, thereby enhancing their work performance levels (Zhou et al., 2024). Therefore, this study posits that employees' direct learning and imitation of artificial intelligence-specific new skills or work behaviors from artificial intelligence will enhance their performance levels. And the higher performance levels achieved through this pathway or the "artificial intelligence-like behaviors" exhibited by employees in subsequent work are referred to as "Mimetic artificial intelligence performance."

In summary, this study proposes the following hypothesis:

H2. Artificial intelligence adoption at work is associated with an increased employees' mimetic artificial intelligence performance by motivating them to learn from artificial intelligence.

2.5. The moderating role of leader's goal orientation

Digital transformation measures adopted by organizations in response to changes in the business environment, such as artificial intelligence adoption, bring changes to the work processes of internal employees while also posing challenges for business leadership. In the context of artificial intelligence adoption at work, leaders' understanding of artificial intelligence technology and goal orientation have a significant impact on employee behavior and performance. If leaders can fully recognize the advantages of artificial intelligence and take appropriate measures (e.g., providing employees with development

resources and other supportive behaviors) to help employees adapt to digital technologies and digital transformation, they can to some extent promote the progress and development of the employees and organization (Yin et al., 2024a). Social influence theory suggests that individuals' psychological and behavioral patterns may change to varying degrees under the influence of others, with the extent of change primarily determined by the importance and proximity of those others (Kelman, 1958). Leaders typically maintain long-term close relationships with employees, assign work tasks, and set performance requirements. Therefore, this study posits that leaders' goal orientations can to some extent alter employees' behavioral tendencies. Building on previous discussions of employees' two distinct behavioral tendencies toward artificial intelligence (utilizing artificial intelligence and learning from artificial intelligence), this study further categorizes leaders' goal orientations into task-oriented and knowledge-oriented approaches to examine their effect on employees' behavioral patterns and performance.

On one hand, task-oriented leaders focus their efforts on reasonably assessing and allocating employees' capabilities and roles to ensure the efficient completion of work tasks (Rüzgar, 2018). They do not overly focus on other aspects unrelated to work tasks, such as employees' personal development. When facing crises or uncertain new environments (e.g., artificial intelligence adoption at work), task-oriented leaders may focus more on the role of artificial intelligence technology in information collection and data analysis to enhance employee work efficiency. They may guide employees to fully recognize and understand the auxiliary functions of artificial intelligence as an assistant and train employees to utilize artificial intelligence technology to address changing work tasks. Combining social influence theory, this study suggests that task-oriented leaders encourage employees to place greater emphasis on the completion of their work tasks and are more inclined to utilize artificial intelligence to enhance their efficiency in addressing work tasks. Therefore, under the leadership of task-oriented leaders, employees may more frequently use artificial intelligence to solve work problems and fully experience the significant improvement in their work efficiency brought by artificial intelligence assistants (Wijayati et al., 2022). As a result, employees may have greater confidence in using artificial intelligence to solve work tasks and problems, i.e., higher problem-solving efficacy.

Additionally, task-oriented leaders encourage employees to work within their assigned responsibilities and scope, and through continuous supervision and timely guidance, ensure that underperforming employees promptly correct inappropriate work behaviors or methods to meet changing task requirements (Ahmed et al., 2024). Therefore, under the guidance and requirements of task-oriented leaders, employees will more actively utilize artificial intelligence to adjust their work behaviors and invest more resources to address changes in task demands caused by external environmental changes, thereby becoming more efficient performers and demonstrating higher adaptive performance.

In summary, this study proposes the following hypotheses:

H3. Leader's goal orientation moderates the effect of artificial intelligence adoption at work on employees' problem-solving efficacy: Task-oriented leaders will strengthen the positive effect of artificial intelligence on employees' problem-solving efficacy.

H4. In the pathway of artificial intelligence adoption at work → problem-solving efficacy → adaptive performance, the mediating role of problem-solving efficacy is moderated by the leader's goal orientation: when the leader's goal orientation is task-oriented, the mediating effect of problem-solving efficacy on the relationship between artificial intelligence adoption at work and adaptive performance is stronger.

On the other hand, knowledge-oriented leaders place particular emphasis on knowledge management within the organization, intentionally implementing various measures to promote the sharing, dissemination, and integration of knowledge within the organization

(González-Mohino et al., 2024). Artificial intelligence provides employees with more time and opportunities to learn (Li et al., 2024), and also engages in self-learning and improvement through interaction with employees, thereby continuously generating new knowledge. Therefore, when knowledge-oriented leaders introduce artificial intelligence into the organization, it may be viewed by leaders as an effective means to assist employees in learning and acquiring new knowledge (Abbas et al., 2026). Unlike task-oriented leaders, knowledge-oriented leaders are more supportive of employees' self-improvement and development, and intentionally provide resources and opportunities through artificial intelligence to empower employees, promote organizational knowledge flow, and support employee learning (Men and Jia, 2021). Research by Dorta-Afonso et al. (2024) also indicated that under the leadership of knowledge-oriented leaders, employees are more actively engaged in organizational knowledge-sharing activities and exhibit corresponding learning behaviors.

Additionally, knowledge-oriented leaders exhibit higher tolerance for employee errors, avoiding shaming or punishing employees for mistakes made while learning from artificial intelligence (Donate et al., 2022). This suggests that employees with a desire for self-improvement may experience higher psychological safety in a knowledge-oriented leadership environment (Men and Jia, 2021), be more proactive in learning from artificial intelligence, and subsequently exhibit more artificial intelligence-like behaviors in their work, resulting in higher mimetic artificial intelligence performance. Based on this, this study argues that, compared to task-oriented leaders, knowledge-oriented leaders enhance employees' willingness to engage in upward assimilation comparisons with artificial intelligence and learn from artificial intelligence to achieve self-improvement. Therefore, this study proposes the following hypotheses:

H5. Leader's goal orientation moderates the impact of artificial intelligence adoption at work on employees' problem-solving efficacy: Knowledge-oriented leaders will strengthen the positive effect of artificial intelligence adoption at work on employees' learning from artificial intelligence.

H6. In the pathway of artificial intelligence adoption at work → learning from artificial intelligence → mimetic artificial intelligence performance, the mediating role of learning from artificial intelligence is moderated by leader goal orientation: When the leader's goal orientation is knowledge-oriented, the mediating effect of learning from artificial intelligence on mimetic artificial intelligence performance is stronger.

In light of the above discussions, the research model is presented in Fig. 1.

3. Research design

3.1. Development of the mimetic artificial intelligence performance scale

The scale development section of this study draws on the recommendations and processes outlined by Spector (1992) and Li et al. (2024) regarding scale development.

3.1.1. Variable definition

Employee performance refers to the behavioral activities employees undertake to complete work tasks or achieve specific objectives, which can be observed and quantified. Many scholars view employee performance as a multidimensional construct. For example, Borman and Motowidlo (1997) categorized employee performance into task performance and contextual performance. Pulakos et al. (2000) further divided performance into task performance, contextual performance, and adaptive performance. In previous discussions, we note that adaptive performance refers to the adaptive behaviors exhibited by employees (Shoss et al., 2012). And adaptive performance has become a critical factor for employees and organizations to survive and thrive in turbulent environments. In the face of the displacement threat posed by artificial intelligence adoption at work, employees enhance their problem-solving efficacy by leveraging artificial intelligence to ensure stable performance in turbulent environments. From this perspective, adaptive performance is essentially a relatively passive behavioral activity, meaning that the primary driver of adaptive behavior is the potential threat posed by organizational environmental changes rather than employees' proactive willingness to transform themselves. Conversely, proactive performance is an employee-initiated, future-oriented behavior aimed at changing personal work situations, work roles, or personal characteristics, as well as organizational operational methods (Griffin et al., 2007; Varela-Neira et al., 2018). Unlike adaptive performance, proactive performance emphasizes employees' proactive willingness to identify potential opportunities and issues, transform themselves to better perform their duties, and achieve individual and organizational goals. It represents an active challenge to the current work situation rather than passive adaptation (Fay and Frese, 2001).

This study notes that in the face of potential development opportunities presented by artificial intelligence adoption at work, employees may proactively learn from artificial intelligence to achieve self-development and mimetic artificial intelligence performance, defining mimetic artificial intelligence performance as new skills or work behaviors with artificial intelligence-like characteristics that employees learn from artificial intelligence. Employees learning from artificial intelligence highlights their clear understanding of the future development trends of artificial intelligence within the organization and the

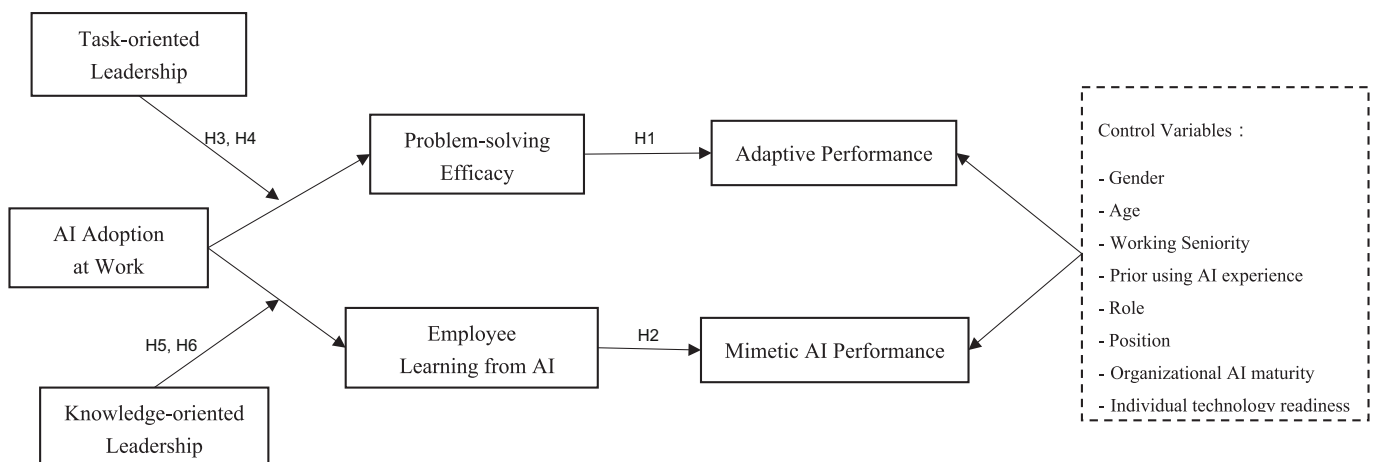


Fig. 1. Research model.

potential opportunities for self-improvement (Li et al., 2024). Thus, learning from artificial intelligence can be regarded as a proactive and forward-thinking transformative behavior (Chen et al., 2023). Mimetic artificial intelligence performance originates from but goes beyond employees' learning from artificial intelligence. The process by which employees acquire mimetic artificial intelligence performance is essentially one of deep observation, imitation, and application of artificial intelligence's skills, followed by internalization. Employees identify artificial intelligence's advantageous dimensions through upward assimilation comparisons, gradually transforming observed artificial intelligence behaviors (directions and informations of progress) into knowledge they can absorb through practical application, and further integrating this into their own capability systems. This learning and internalization process fundamentally distinguishes mimetic artificial intelligence performance from behaviors such as technology adaptation, which remain at a superficial level of employee response or adaptation to new technologies, such as exploitation or avoidance (Bala and Venkatesh, 2016). And compared to learning from artificial intelligence, mimetic artificial intelligence performance indicates that employees adopt a more proactive and in-depth approach toward artificial intelligence. Based on this, this study further defines mimetic artificial intelligence performance as a kind of proactive performance exhibited by employees after actively learning from artificial intelligence, which incorporates artificial intelligence-like characteristics.

3.1.2. Scale design

The scale design for this study was conducted in two phases.

In the first phase, the keyword "employee proactive performance" was searched in academic databases such as WOS (Web of Science database). By reviewing relevant papers, an initial understanding was gained of the connotations, research trends, and measurement perspectives related to "employee proactive performance." The results of the literature review indicated that employee proactive performance stems from related concepts such as proactivity, taking charge, in-role behavior (Griffin et al., 2007; Morrison and Phelps, 1999). It emphasizes that employees adjust their work state and methods through proactive innovation and changes in their work behavior to enhance task performance, exhibiting characteristics such as future orientation, foresight, and innovation (Parker et al., 2006). Researches on proactive performance have shown a trend toward greater refinement in recent years. For example, team proactive performance expands the research focus from individuals to groups, stating that team proactive performance is the result of coordinated efforts among team members (Williams et al., 2010). And proactive customer service performance limits the research context to customer consumption and the research subjects to service personnel, emphasizing that service personnel improve customer service quality through proactive behaviors (such as sharing information with customers, predict customer needs and proactively develop response strategies) to enhance customer service quality (Hur and Shin, 2024). In terms of measuring proactive performance, Varela-Neira et al. (2018) screened nine measurement dimensions proposed by Griffin et al. (2007) for individuals, teams, and organizations in terms of proficiency, adaptivity, and proactivity, such as "Initiated better ways of doing your core tasks" as a proactive performance scale and demonstrated its validity. Lee et al. (2018) selected three dimensions from Morrison and Phelps (1999) targeting individual responsible behavior, such as "Tried to introduce new structures, technologies, or approaches to improve efficiency," as a proactive performance scale and demonstrated its validity. As such, there is a lack of original scales for measuring proactive performance, and scholars' measurement perspectives primarily focus on specific proactive behaviors of individuals and their actual impact on work performance.

In the second phase, this study selected 7 employees and 4 managers from three Chinese internet companies as semi-structured interview subjects. First, based on the results of the literature review, an interview outline was designed (e.g., Do you have the habit of using artificial

intelligence in your work? Do you have the idea of learning from artificial intelligence after using it? Did you apply the logical thinking of generative artificial intelligence to help yourself innovate?). Second, through on-site visits and instant messaging software such as WeChat, a combination of online and offline methods was used to communicate with the interview subjects and systematically record the interview materials. Combining the interview results with Li et al.'s (2024) research on the specific objects and content of employees learning from artificial intelligence, this study measures mimetic artificial intelligence performance from the following three aspects: (1) decision-making behaviors that imitate the predictive and forward-thinking mindset of generative artificial intelligence. For example, marketing department employees imitate generative artificial intelligence decision-making logic and methods by collecting and analyzing historical customer data to generate big data models, predict market or customer demand trends, and determine marketing strategies. (2) Imitating the work optimization behavior of collaborative artificial intelligence robots through real-time feedback and dynamic adjustment mechanisms. For example, artificial intelligence robots adjust and re-plan their routes in real time after encountering obstacles to reach their destination. Employees imitate this dynamic feedback mechanism to monitor, evaluate, and adjust their own work behavior and methods in real time. (3) Imitating the iterative innovation behavior of augmenting artificial intelligence machine learning models. For example, when developing a smart home device, product design employees imitate artificial intelligence's deep iterative learning methods, continuously collect user feedback data based on the initial prototype and after multiple rounds of testing and improvements, ultimately launch a product with more complete functionality.

To ensure the content validity of the scale, authors invited 4 employees from the aforementioned company, 6 experts in management, to evaluate and revise the scale items from two aspects: "whether they can reflect the characteristics and connotations of the construct" and "whether they belong to the corresponding dimension." All participants were clearly informed of the general content of this research and the connotations of the mimetic artificial intelligence performance construct before participating in the scale revision. The revised scale consists of three items, as shown in Table 1. Finally, this study adopted a 5-point Likert scale (strongly disagree—strongly agree) to compile the scale items into a questionnaire for data collection.

3.1.3. Data collection

This study employed an online questionnaire to collect data, which was then analyzed using SPSS 26.0. Screening questions were designed (e.g., Is your company an Internet company? Does your company utilize artificial intelligence tools such as generative artificial intelligence? Have you used artificial intelligence tools in your work? Have you imitated artificial intelligence behavior in your work?) to select participants who met the study criteria. To ensure response rate and quality, each participant who completed the questionnaire received a reward of 2 RMB (approximately US\$0.2788). The survey results showed that a total of 320 questionnaires were distributed, with 13 invalid questionnaires excluded, resulting in 307 valid questionnaires, yielding a valid response rate of 95.93 %. Among them, 176 were male, accounting for 57.32 %. Most participants were aged 26–40, totaling 189, accounting for 61.56 %; 77 were over 40, accounting for 25.08 %; and 41 were aged 18–25, accounting for 13.36 %. Among the respondents, 96 had been working for less than one year, accounting for 31.27 %; the majority had been working for 1–5 years, totaling 175, accounting for 57 %; and 36 had been working for over ten years, accounting for 11.73 %. The majority of respondents had used artificial intelligence for over 6 months, accounting for 72.64 %. In terms of job role, most respondents were first-line employees, totaling 217 people, accounting for 70.68 %; there were 58 front-line managers, accounting for 18.89 %; 22 middle managers, accounting for 7.17 %; and 10 top managers, accounting for 3.26 %. In terms of job position, 46 were service employees, accounting for 14.98

Table 1

Measurement items, reliability and validity of mimetic artificial intelligence performance (SPSS, $N = 307$).

Variable	Item	EFA factor loading	Total variance explained	Cronbach's α
<i>When using artificial intelligence, I will try to apply...</i>				
Mimetic artificial intelligence performance (KMO = 0.719)	The logical thinking of generative artificial intelligence (such as anticipatory thinking) to comprehensively analyze and synthesize issues, thereby improving decision-making quality (MAIP1)	0.856	73.39 %	0.818
	The behavioral dynamic feedback adjustment mechanism of collaborative robots, monitoring and modifying my own work style to improve the work efficiency (MAIP2)	0.859		
	The machine learning model of augmenting artificial intelligence, deeply iterating and optimizing innovative content to improve innovation efficiency (MAIP3)	0.855		

%; 103 were technical employees, accounting for 33.55 %; 68 were marketing and planning employees, accounting for 22.15 %; and 90 were managers, accounting for 29.32 %.

3.1.4. Item analysis

The purpose of item analysis is to verify whether the scale can distinguish between participants of different levels. In this study, the Likert scores of the three items in each questionnaire were summed separately, and the participants were ranked in order of total score. The top 27 % of total scores were set as the high-score group (total score ≥ 12), and the bottom 27 % of total scores were set as the low-score group (total score ≤ 8). A t -test was conducted to compare the mean scores of each item between the high- and low-score groups, verifying whether there were significant differences in item scores between the two groups. The analysis results showed that the t -values for all items in the scale were greater than the critical value of 3 and reached the level of significance ($p < 0.05$), indicating that the items have good discriminant validity. Additionally, the r -value analysis showed that the correlation coefficients between the scale items and the total scale score were all greater than 0.8 and reached the level of significance ($p < 0.01$), further validating that the items have high discriminant validity.

3.1.5. Reliability and validity analysis

As shown in Table 1, the Cronbach's α value for the mimetic artificial intelligence performance variable is 0.818, indicating that the scale has high reliability. Furthermore, the sample was subjected to Kaiser-Meyer-Olkin (KMO) and Bartlett's sphericity tests. The results showed that the KMO value was 0.719, and the Bartlett's sphericity test value was 318.729, both reaching a significant level ($p < 0.001$). This indicates that the scale can undergo factor analysis. Next, principal component analysis was used to extract factors with eigenvalues greater than 1, and factor rotation was performed using the varimax rotation method. A single factor with an eigenvalue greater than 1 was extracted.

Additionally, the scree plot showed that introducing a second factor slowed the overall trend of change, so the factor prior to the inflection point could be retained (Li et al., 2024). The total variance explained was 73.393 %, exceeding 60 %. Factor loadings of the items ranged from 0.855 to 0.859, and the commonality sizes ranged from 0.731 to 0.737, indicating that all three items of the scale have good convergent validity.

3.2. Three-wave experience sampling study

3.2.1. Participants and experimental procedures

The three-wave experience sampling study method is a high-frequency tracking questionnaire survey method. Data collected using this method can reflect the dynamic changes of individual participants and has long-term effects. The specific experimental steps are as follows.

First, design the questionnaire. Two experts in management used the back-translation method to translate portions of a mature scale developed by foreign scholars into Chinese items. Prior to the centralized distribution of the questionnaire, researchers recruited 15 business administration students for a pre-test. All participants indicated that the Chinese-language items were readable and faithful to the original English meaning. Second, an online questionnaire was generated using the online questionnaire platform (<https://www.wjx.cn>). Participants who completed all three waves of the survey were offered a reward of 15 RMB (approximately US\$2.089). Third, recruit participants. This study invited 8 Chinese internet companies as research subjects and contacted relevant company managers to introduce the theme, significance, and requirements of the study. Following online communication, the 8 companies provided a total of 374 lists of personnel (employees and leaders) eligible to participate in the experiment, which were then numbered and anonymized by the researchers. Fourth, distribute the questionnaires. At Time Point 1 (T1, May 2025), distribute questionnaires measuring participants' basic information, artificial intelligence adoption at work, and leader's orientation (for a response rate of 91.98 %); at Time Point 2 (T2), a questionnaire measuring problem-solving efficacy and employee learning from artificial intelligence was distributed to employees who had completed the T1 survey (for a response rate of 93.90 %); at Time Point 3 (T3), employees who had completed the T1 and T2 surveys were given questionnaires to complete on adaptive performance and mimetic artificial intelligence performance (for a response rate of 95.98 %). Each survey wave was spaced two weeks apart. Respondents were required to complete each questionnaire within 48 h of distribution. Fifth, data analysis. After excluding 2 invalid questionnaires, a sample of 308 responses was obtained (for a final response rate of 82.35 %). Detailed information is shown in Table 2.

3.2.2. Variable measurement

Each item in the questionnaire was measured using a 5-point Likert scale (1 represents strongly disagree, 5 represents strongly agree). Some items in the questionnaire were adjusted to better suit the context of this study.

1. The artificial intelligence adoption at work (AIA, T1) scale is adapted from Polisetty et al. (2024), which includes 4 items, such as "When using artificial intelligence, I think that artificial intelligence is compatible with most aspects of my work."
2. The Problem-solving efficacy (PSE, T2) scale is adapted from Lin (2023), which includes 4 items, such as "I feel confident when solving problems with artificial intelligence in the organization."
3. To assess employees' adaptive performance (AP, T3), we selected 6 items from the individual-level adaptive performance scale developed by Marques-Quinteiro et al. (2015). For example, "When using artificial intelligence, I can find innovative ways to deal with unexpected events."
4. The Employee Learning from Artificial Intelligence (LAI, T2) scale is adapted from Li et al. (2024), which includes 3 items, such as "When

Table 2
Sample's sprofile (N = 308).

Variables	Categories	Frequency	Percentage
Gender	Male	207	67.21 %
	Female	101	32.79 %
Age	18–25	65	21.10 %
	26–35	127	41.24 %
	36–40	93	30.19 %
	older than 40	23	7.47 %
Working Seniority	0–3 years	112	36.36 %
	3–5 years	104	33.77 %
	5–10 years	65	21.10 %
	More than 10 years	27	8.77 %
Prior using artificial intelligence experience	0–3 months	24	7.79 %
	3–6 months	74	24.03 %
	6 months–1 year	114	37.01 %
	More than 1 years	96	31.17 %
Role	First-line employees	243	78.90 %
	First-line managers	24	7.79 %
	Middle managers	23	7.47 %
	Top managers	18	5.84 %
Position	Service employees	28	9.09 %
	Technical employees	132	42.86 %
	Marketing planning employees	83	26.95 %
	Managers	65	21.10 %
Organizational artificial intelligence maturity	Basic tool application stage	32	10.39 %
	Highly intelligent stage	267	86.69 %
	Strategic-driven R&D stage	9	2.92 %
Individual technology readiness	Hard to accept	26	8.44 %
	Utilize artificial intelligence with ease	203	65.91 %
	Innovatively apply artificial intelligence	79	25.65 %

using generative artificial intelligence machines, I learn the logical thinking behind artificial intelligence generation schemes.”

- The Mimetic Artificial Intelligence Performance (MAIP, T3) variable is measured using a scale developed by this study, as described in Table 1.
- The Task-oriented leadership (TOL, T1) scale is adapted from Döring et al. (2024) and includes 5 items, such as “My leader works to ensure a high professional quality of employees' performance.”
- The Knowledge-oriented leadership (KOL, T1) scale is adapted from Donate and de Pablo (2015) and includes 6 items, such as “My leader promotes the acquisition of external knowledge.”
- Control variables (T1). The effects of artificial intelligence adoption at work may vary due to individual differences (Tong et al., 2021). Therefore, in addition to using gender and age as basic control variables, this study additionally included factors such as working seniority, prior using artificial intelligence experience as control variables to explore their potential role in the mechanism by which artificial intelligence adoption at work influence work performance.

3.2.3. Scale reliability and validity

First, exploratory and confirmatory factor analysis was conducted on the scales used in this study using SPSS 26.0 and AMOS 26.0. The results are shown in Table 3. In the exploratory factor analysis, the minimum total variance explained by all variables was 62.195 % (AP), exceeding 60 %; factor loadings range from 0.764 to 0.871, demonstrating good convergent validity. The confirmatory factor analysis results indicates that the 7-factor model has better fit than other competing models and is the optimal model. Additionally, in the seven-factor model, all variable item factor loadings range from 0.624 to 0.840, all exceeding 0.5. Furthermore, Cronbach's α and CR values are both greater than 0.7, and AVE values are all greater than 0.5, indicating that the scales used in this

Table 3
The reliability and validity of variables (SPSS and AMOS, N = 308).

Variables	Item	EFA loadings	Total variance explained	CFA loadings (7-factor model)	AVE	CR	Cronbach's α
Artificial intelligence adoption at work (KMO = 0.791)	AIA1	0.835	65.270 %	0.774	0.539	0.824	0.822
	AIA2	0.785		0.700			
	AIA3	0.824		0.757			
	AIA4	0.787		0.703			
Problem-solving efficacy (KMO = 0.803)	PSE1	0.862	66.597 %	0.816	0.558	0.834	0.833
	PSE2	0.797		0.704			
	PSE3	0.814		0.753			
	PSE4	0.791		0.709			
Adaptive performance (KMO = 0.898)	AP1	0.808	62.195 %	0.727	0.547	0.879	0.878
	AP2	0.764		0.709			
	AP3	0.785		0.742			
	AP4	0.800		0.752			
	AP5	0.808		0.760			
	AP6	0.796		0.747			
Employee learning from artificial intelligence (KMO = 0.684)	LAI1	0.846	68.100 %	0.786	0.528	0.767	0.765
	LAI2	0.847		0.759			
	LAI3	0.781		0.624			
Mimetic artificial intelligence performance (KMO = 0.703)	MAIP1	0.851	69.978 %	0.745	0.549	0.785	0.785
	MAIP2	0.822		0.695			
	MAIP3	0.836		0.781			
Task-oriented leadership (KMO = 0.881)	TOL1	0.847	71.988 %	0.805	0.650	0.903	0.902
	TOL2	0.828		0.777			
	TOL3	0.861		0.839			
	TOL4	0.871		0.831			
	TOL5	0.834		0.778			
Knowledge-oriented leadership (KMO = 0.927)	KOL1	0.846	71.823 %	0.800	0.661	0.921	0.921
	KOL2	0.862		0.840			
	KOL3	0.847		0.817			
	KOL4	0.852		0.812			
	KOL5	0.846		0.822			
	KOL6	0.831		0.787			

Note: AIA: Artificial intelligence adoption at work; PSE: Problem-solving efficacy; AP: Adaptive performance; LAI: Employee learning from artificial intelligence; MAIP: Mimetic artificial intelligence performance.

TOL: Task-oriented leadership; KOL: Knowledge-oriented leadership. Same as below.

study have good reliability and validity.

3.2.4. Common method bias testing

Since the data for both the mediating variables and outcome variables in the research questionnaire were sourced from colleagues, there is a phenomenon of data homogeneity, which may introduce common method bias from the same data source. This study employed the following three methods to test for the presence of common method bias.

First, the Harman single-factor test method. Using SPSS 26.0, factor analysis was conducted on all observed variables using latent variables. The results identify 6 factors, with the largest factor explaining 33.644 % of the variance (below 40 %), indicating no obvious common method bias. Second, confirmatory factor analysis with a single factor. As shown in the Table 4, the 1-factor model fits significantly worse than the seven-factor model, indicating no obvious common method bias. Third, confirmatory factor analysis with a common method factor. A common method factor was added to the 7-factor model, as shown in Table 4. Compared to the original 7-factor model, the 7-factor model with the common method factor has RMSEA and SRMR differences both less than 0.5, CFI and TLI differences both less than 0.1. Therefore, there is no unmeasured common method bias. Additionally, this study implemented various procedural measures during the questionnaire survey to reduce the impact of common method bias, such as using simple and understandable language to translate the scales, anonymizing participants, keeping questionnaire data confidential, and arranging all exogenous variable measurement items before endogenous variable items in the questionnaire design (Wong et al., 2024).

4. Empirical results

4.1. Data analysis strategy

This study employs Partial Least Squares Structural Equation Modeling (PLS-SEM) and Artificial Neural Networking (ANN) to capture both linear and nonlinear relationships among model variables.

The reasons for choosing PLS-SEM over traditional CB-SEM are as follows: first, this study attempts to introduce social comparison theory into HAI scenarios while introducing a new variable (mimetic artificial intelligence performance). Thus, this study is more oriented toward prediction and theoretical development. And PLS-SEM is perfectly suited to investigate predictive models (ie. through the blindfolding-based Q^2 statistic and Cross-validated Predictive Ability Test), as the method strikes a balance between machine learning methods, which are fully predictive in nature, and CB-SEM, which focuses on confirmation and model fit (Richter et al., 2016). Most scholars also recommend using PLS-SEM for model testing involving theoretical development and new variables (e.g., Hair et al., 2017; Sarstedt et al., 2022). Second, the model in this study includes 7 main variables, making it relatively complex; and the one-sample Kolmogorov–Smirnov test shows that all p -values for the indicator construct are less than 0.05, indicating that the data distribution is non-normal. Therefore, for the sample in this study ($N = 308$), using bootstrapping based on PLS-SEM may produce a smaller average bias than CB-SEM (based on the maximum likelihood method) (Hair et al., 2012; Sharma and Kim, 2013; Tew et al., 2021).

Third, PLS-SEM, as an independent structural equation modeling method, does not require secondary corrections like CB-SEM (Zatori et al., 2018). And the main reason for combining PLS-SEM and ANN for hypothesis testing is that while PLS-SEM can predict linear relationships between variables, it cannot effectively capture nonlinear relationships in the model (Lim et al., 2022). ANN operates based on causal relationships, enabling the detection of nonlinear and non-compensatory relationships within the model while exhibiting strong robustness and adaptability. It also does not require data to follow a normal distribution (Wong et al., 2024). Thus, combining the two methods can effectively detect both linear and nonlinear relationships within the model, enhance predictive accuracy, and make the analysis results more aligned with real-world conditions.

4.2. Assessing the outer measurement model

The results are shown in Table 5. It can be seen that the values of various indicators calculated using Smartpls 4 differ slightly from those in Table 3, but overall they are still acceptable. For example, Cronbach's α ranges from 0.765 to 0.922, CR values range from 0.865 to 0.938, and AVE values are all greater than 0.5, which again demonstrates that the scale has good reliability and validity. Table 6 displays the discriminant validity of the constructs, based on the Fornell-Larcker and HTMT criteria. According to the Fornell-Larcker criterion, the square root of each variable's AVE is greater than its correlations with other variables. Moreover, when evaluating the HTMT ratios, the highest ratio is 0.779 (between KOL and TOL), which is below the threshold of 0.85. Overall, the constructs exhibit strong discriminant validity.

4.3. Structural model path coefficient test

The standard bootstrap method (10,000 bootstrap samples) was used to test the significance of the path coefficients. The results of the structural model path coefficient tests are shown in Fig. 2 and Table 7. Artificial intelligence adoption at work (AIA) has a significant positive effect on adaptive performance (AP) through problem-solving efficacy (PSE) ($\beta = 0.180, p < 0.001$), supporting H1; simultaneously, AIA has a significant positive effect on mimetic artificial intelligence performance through learn from artificial intelligence (LAI) ($\beta = 0.127, p < 0.01$), supporting H2. Furthermore, the interaction term between task-oriented leadership (TOL) and AIA has a significant positive effect on PSE ($\beta = 0.365, p < 0.001$), thus supporting H3; the interaction term between knowledge-oriented leadership (KOL) and AIA has a significant positive effect on LAI ($\beta = 0.230, p < 0.001$), thus supporting H5. To more intuitively demonstrate the moderating effects of leader's orientation, moderation effect diagrams were generated using Smartpls 4, as shown in Figs. 3 and 4. It can be seen that compared to low task-oriented leadership, the regression line slope is steeper for high task-oriented leadership; compared to knowledge-oriented leadership, the regression line slope is steeper for high knowledge-oriented leadership, further supporting H3 and H5.

Table 7 further indicates that under any level grouping, the indirect effect of AIA on AP does not include 0 in the 95 % confidence interval; and the index of conditional mediation effect is 0.117, with a confidence

Table 4
Confirmatory factor analysis results (AMOS, $N = 308$).

Models	Factors	χ^2/df	RMSEA	SRMR	CFI	TLI	IFI	PCFI
7-factor model +common method factor		1.328	0.033	0.0345	0.976	0.971	0.976	0.802
7-factor model	AIA, PSE, AP, LAI, MAIP, TOL, KOL	1.425	0.037	0.0365	0.966	0.962	0.967	0.858
6-factor model	AIA + PSE, AP, LAI, MAIP, TOL, KOL	2.196	0.062	0.0557	0.903	0.893	0.904	0.814
5-factor model	AIA + PSE+ AP, LAI, MAIP, TOL, KOL	2.584	0.072	0.0626	0.870	0.858	0.872	0.794
4-factor model	AIA + PSE + AP+ LAI, MAIP, TOL, KOL	2.927	0.079	0.0694	0.841	0.827	0.842	0.774
3-factor model	AIA + PSE + AP + LAI + MAIP, TOL, KOL	3.250	0.086	0.0728	0.813	0.798	0.814	0.754
2-factor model	AIA + PSE + AP + LAI + MAIP+TOL, KOL	5.600	0.122	0.1191	0.616	0.588	0.618	0.574
1-factor model	AIA + PSE + AP + LAI + MAIP+TOL+ KOL	6.240	0.131	0.1289	0.562	0.530	0.564	0.524

Table 5

Construct reliability and convergent validity (Smartpls, N = 308).

Latent variable/Indicators	Loadings	t-values	p-values	Cronbach's α	rho_a	rho_c	AVE
Artificial intelligence adoption at work (AIA)				0.822	0.823	0.883	0.653
AIA1	0.835	30.561	0				
AIA2	0.786	24.15	0				
AIA3	0.822	38.061	0				
AIA4	0.788	26.338	0				
Problem-solving efficacy (PSE)				0.832	0.835	0.888	0.666
PSE1	0.86	50.476	0				
PSE2	0.788	24.575	0				
PSE3	0.809	30.565	0				
PSE4	0.805	27.292	0				
Adaptive performance (AP)				0.878	0.880	0.908	0.622
AP1	0.781	27.632	0				
AP2	0.759	22.352	0				
AP3	0.796	30.902	0				
AP4	0.799	32.043	0				
AP5	0.8	31.515	0				
AP6	0.794	29.83	0				
Employee learning from artificial intelligence (LAI)				0.765	0.769	0.865	0.681
LAI1	0.853	37.725	0				
LAI2	0.839	33.433	0				
LAI3	0.782	20.21	0				
Mimetic artificial intelligence performance (MAIP)				0.785	0.793	0.874	0.699
MAIP1	0.851	37.195	0				
MAIP2	0.804	25.714	0				
MAIP3	0.853	39.687	0				
Task-oriented leadership (TOL)				0.903	0.906	0.928	0.720
TOL1	0.85	36.853	0				
TOL2	0.819	27.968	0				
TOL3	0.865	45.867	0				
TOL4	0.867	38.311	0				
TOL5	0.839	31.582	0				
Knowledge-oriented leadership (KOL)				0.922	0.929	0.938	0.717
KOL1	0.828	32.28	0				
KOL2	0.862	45.529	0				
KOL3	0.857	44.244	0				
KOL4	0.853	42.158	0				
KOL5	0.84	36.152	0				
KOL6	0.841	35.057	0				

Table 6

Discriminant validity results for Fornell-Larcker and Heterotrait-Monotrait (HTMT) criteria.

	AIA	AP	KOL	LAI	MAIP	PSE	TOL
AIA	0.808	0.551***	0.291***	0.475***	0.392***	0.362***	0.394***
AP	0.646	0.789	0.348***	0.429***	0.400***	0.491***	0.446***
KOL	0.334	0.388	0.847	0.319***	0.286***	0.256***	0.713***
LAI	0.598	0.525	0.372	0.825	0.374***	0.312***	0.444***
MAIP	0.486	0.482	0.329	0.480	0.836	0.520***	0.284***
PSE	0.437	0.573	0.288	0.390	0.640	0.816	0.280***
TOL	0.457	0.500	0.779	0.531	0.333	0.320	0.848

Note: *** $p < 0.001$. Italic values represent the square roots of AVEs. The upper values represent the squared correlations. The below values represent HTMT ratios.

interval of [0.067, 0.182], indicating that TOL significantly moderates the mediating effect between AIA and AP, and H4 is supported. Similarly, the confidence intervals for the mediating effects of knowledge-oriented leadership on the relationship between AIA and MAIP also do not include 0 at any level grouping. The index of conditional mediation effect is 0.054, with a confidence interval of [0.017, 0.107], which also does not include 0, indicating that KOL significantly moderates the mediating effect between AIA and MAIP, supporting H6.

Effect size (f^2) provides evidence of an exogenous construct's relative impact on an endogenous construct in terms of R^2 changes (Henseler et al., 2016). And the intensity is represented by small, medium and large effects for values above 0.02, 0.15 and 0.35, respectively (Cohen, 1988). As shown in Table 8, the research model average f^2 is 0.204 which represents medium effect of predictor latent variables at the structural level. At the same time, AIA has a greater effect on AP and MAIP than LAI and PSE have on them, indicating that AIA not only has an indirect impact through its effect on LAI and PSE, but its direct effect

on AP and MAIP may be even more significant. Therefore, in organizational management, directly promoting artificial intelligence application may have a more prominent and substantial impact on employees' ultimate performance outcomes. The highest effect is identified of AIA on LAI ($f^2 = 0.337$), indicating that artificial intelligence adoption at work is a particularly impactful driver of employees learning from artificial intelligence and effect of dropping it out of the model will be the highest (Sternad Zabukovšek et al., 2019). Based on this, organizations aiming to enhance such learning behavior should prioritize initiatives promoting artificial intelligence adoption (such as, artificial intelligence training or fostering an artificial intelligence-accepting culture) since this is where resources can be most effectively allocated to drive meaningful change, unlike statistical significance which merely confirms a relationship exists.

Additionally, as shown in Table 9, among the control variables, only gender (Gender $\rightarrow$ AP, $\beta = 0.16$, $p < 0.05$) and age (Age $\rightarrow$ MAIP, $\beta = 0.089$, $p < 0.05$) have significant effects, while the remaining variables

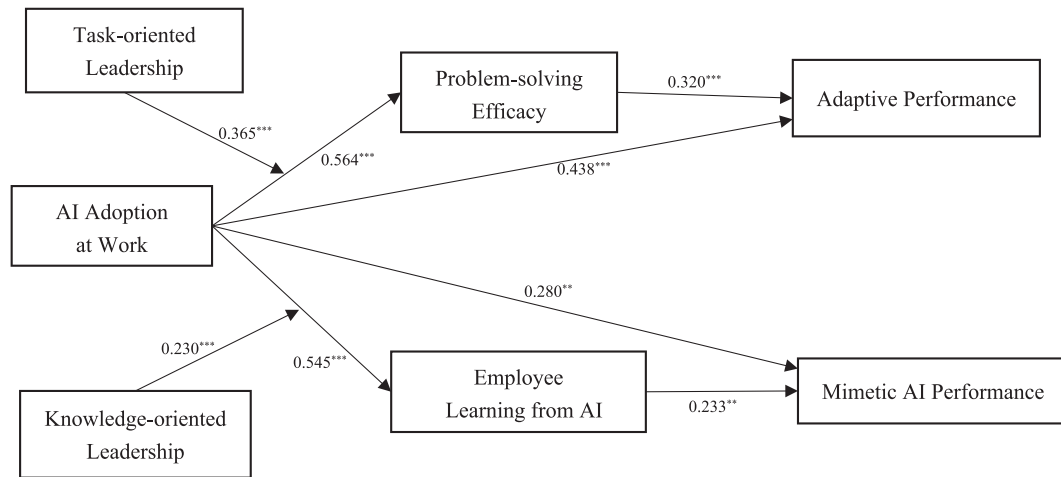


Fig. 2. Direct effects (Main factors).

Table 7
Hypotheses testing results.

Hypotheses	Paths	β	Standard deviation (STDEV)	t-values	LCI 5 %	UCI 95 %	Decision
	Mediating effects						
H1	AIA -> PSE- > AP	0.180***	0.046	3.921	0.113	0.265	Supported
H2	AIA -> LAI- > MAIP	0.127**	0.046	2.736	0.048	0.203	Supported
	Moderating effects						
H3	TOL x AIA -> PSE	0.365***	0.06	6.054	0.265	0.463	Supported
H5	KOL x AIA -> LAI	0.230***	0.067	3.449	0.121	0.34	Supported
	Conditional Mediated Effect						
	Moderating variable: TOL						
H4	AIA -> PSE -> AP (conditional on TOL at +1 SD)	0.299***	0.073	4.122	0.187	0.427	Supported
	AIA -> PSE -> AP (conditional on TOL at mean)	0.182***	0.045	4.085	0.117	0.265	
	AIA -> PSE -> AP (conditional on TOL at −1 SD)	0.064*	0.032	2.038	0.023	0.129	
	Index of conditional mediation effect	0.117***	0.034	3.413	0.067	0.182	
	Moderating variable: KOL						
H6	AIA -> LAI -> MAIP (conditional on KOL at +1 SD)	0.170*	0.067	2.515	0.064	0.288	Supported
	AIA -> LAI -> MAIP (conditional on KOL at mean)	0.119**	0.045	2.643	0.046	0.197	
	AIA -> LAI -> MAIP (conditional on KOL at −1 SD)	0.068*	0.03	2.291	0.03	0.133	
	Index of conditional mediation effect	0.054*	0.027	1.979	0.017	0.107	

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The 95 % confidence intervals are the Bias-corrected confidence intervals.

did not have a significant impact on the dependent variables.

4.4. The explanatory power and predictive relevance of the structural model

The R^2 and R^2 adjusted values for all endogenous variables are shown in Table 10. At the same time, it can be seen that the SRMR (Standardized Root Mean Square Residual) value of the model is 0.042, which is lower than the critical value of 0.08, indicating that the model has good fit validity. Furthermore, the Stone-Geisser's Q^2 values for all endogenous variables range from 0.165 to 0.328, indicating that the model has high predictive relevance (Yuan et al., 2022).

4.5. CVPAT for out-of-sample predictive validity

Although the model in this study demonstrated good fit and predictive power on the sample data, its predictive ability and general applicability to out-of-sample data need to be further tested using CVPAT (Cross-validated Predictive Ability Test) (Sharma et al., 2023).

As shown in Table 11, in the two sets of CVPAT results for the PLS-SEM model compared with the Indicator Average (IA) and the PLS-SEM model compared with the Linear (LM) model, the average loss difference for the Overall model was negative. This indicates that the average predictive loss of the PLS-SEM model is significantly lower than that of the IA and LM benchmark predictive models. Additionally, since the obtained p -values are all less than 0.05, this suggests that the predictive capability of the PLS-SEM model is significantly superior to that of the aforementioned two benchmark models. Therefore, the model proposed in this study demonstrates good predictive capability and practical applicability.

4.6. Artificial neural network analysis

As mentioned earlier, PLS-SEM can effectively analyze linear relationships in a model but has difficulty capturing nonlinear relationships. ANN can handle both linear and nonlinear relationships, but due to the ANN's "black-box" nature, an ANN approach alone is not adequate for hypothesis testing (Lee et al., 2016; Suhail et al., 2024). By

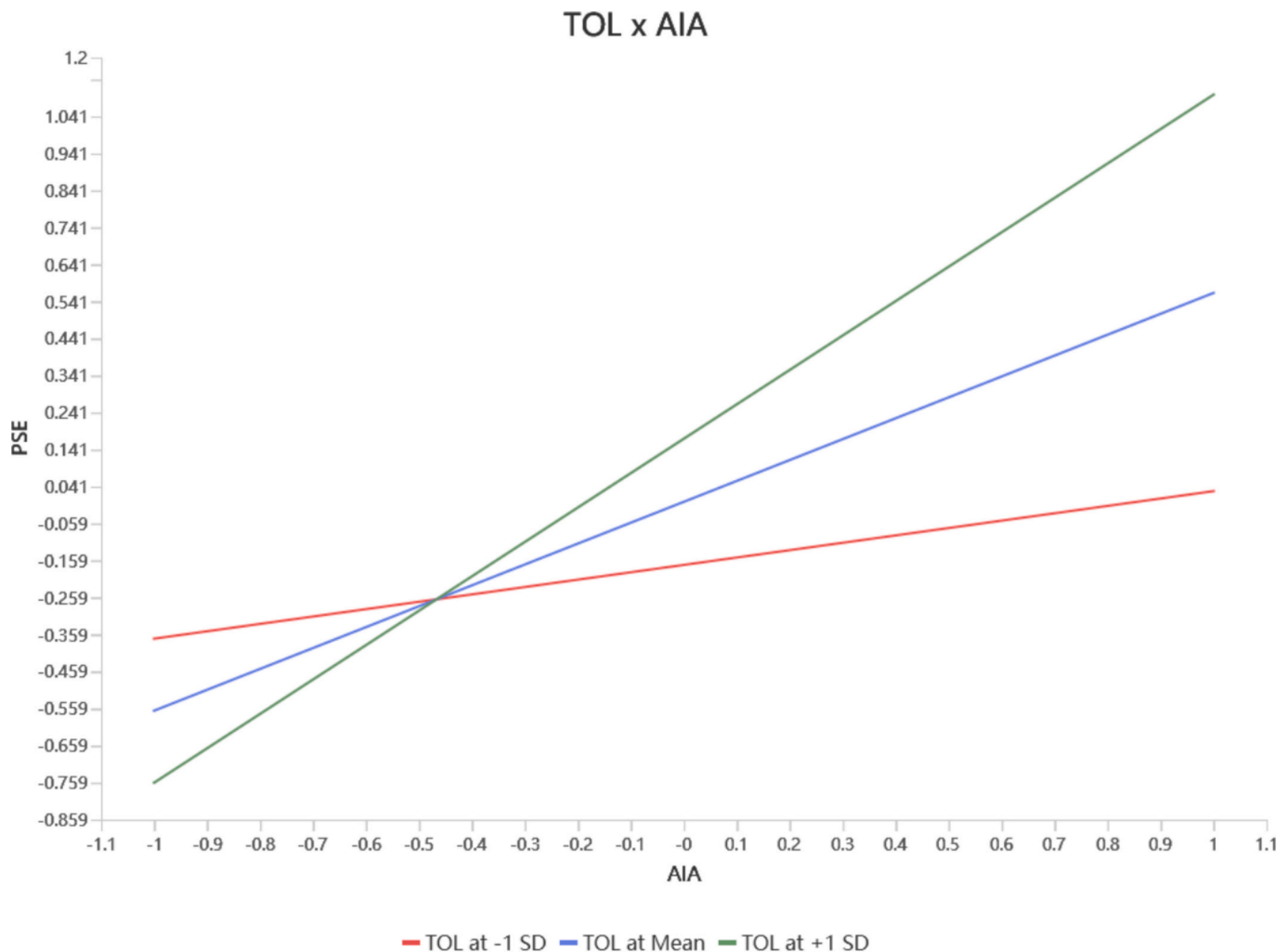


Fig. 3. Simple slope for moderating effect of TOL on AIA and PSE relation.

integrating PLS-SEM into ANN, it not only enhances the interpretability of network connection weights in ANN models but also compensates for the shortcomings of PLS-SEM methods by leveraging the nonlinear fitting ability of ANN models, thereby making the analysis results more reflective of reality (Wong et al., 2024). Based on this, this study used SPSS 26.0 to construct 4 separate neural network models (Models A, B, C, and D) for endogenous constructs (AP, MAIP, PSE, and LAI). According to Table 7, all main factors (AIA, PSE, LAI) are significant and can be used as inputs to the ANN (Arpaci et al., 2022). The structure of ANN models has two hidden layers and the sigmoid function was selected as the activation function for the hidden and output layers (Kalinić et al., 2021; Suhail et al., 2024). Ten-fold cross-validation was employed to address potential overfitting, with 10 % of the data used for testing and the remaining 90 % for training (Al-Sharafi et al., 2023). Additionally, to increase the effectiveness of training, i.e. to provide shorter training times and better performance, all inputs and outputs were normalized to the range [0,1] (Liébana-Cabanillas et al., 2017).

First, the RMSE value for each model was calculated to assess predictive accuracy. As shown in Table 12, the average RMSE values of all ANN models range from 0.0129 to 0.0172 for training sets and from 0.0146 to 0.0167 for testing sets, indicating strong predictive accuracy of the neural network models (Teo et al., 2015). Second, Table 13 presents the sensitivity analysis of the neural network models, which evaluates the relative importance of exogenous variables in predicting endogenous variables. In Model A, artificial intelligence adoption at work is more influential than problem-solving efficacy in predicting

adaptive performance. In Model B, artificial intelligence adoption at work is more important than employee learning from artificial intelligence in predicting mimetic artificial intelligence performance. Despite being a single-neuron model, Models C and D show a normalized importance of 100 %. Third, as shown in Table 14, results from PLS-SEM and ANN were compared using path coefficients and normalized relative importance (Wong et al., 2024). The findings are full consistent which validates the robustness of the empirical results from the first PLS-SEM analysis stage.

5. General discussion

The core argument of this study is that artificial intelligence adoption at work has dual effects on employees. Based on this, employees regulate such effects through social comparison with artificial intelligence while improving their own performance (adaptive performance and mimetic artificial intelligence performance). Therefore, the model and conclusions of this study are applicable to employees who can clearly perceive the changes brought about by artificial intelligence adoption at work and are willing to actively adjust their performance to adapt to them. For employees who exhibit strong trust and dependence on artificial intelligence upon exposure to it, they may not perceive the negative impacts or development opportunities of artificial intelligence on themselves, and thus may not engage in social comparison behavior with it. Taking artificial intelligence hallucinations as an example, when artificial intelligence generates content that contradicts facts, deviates from the

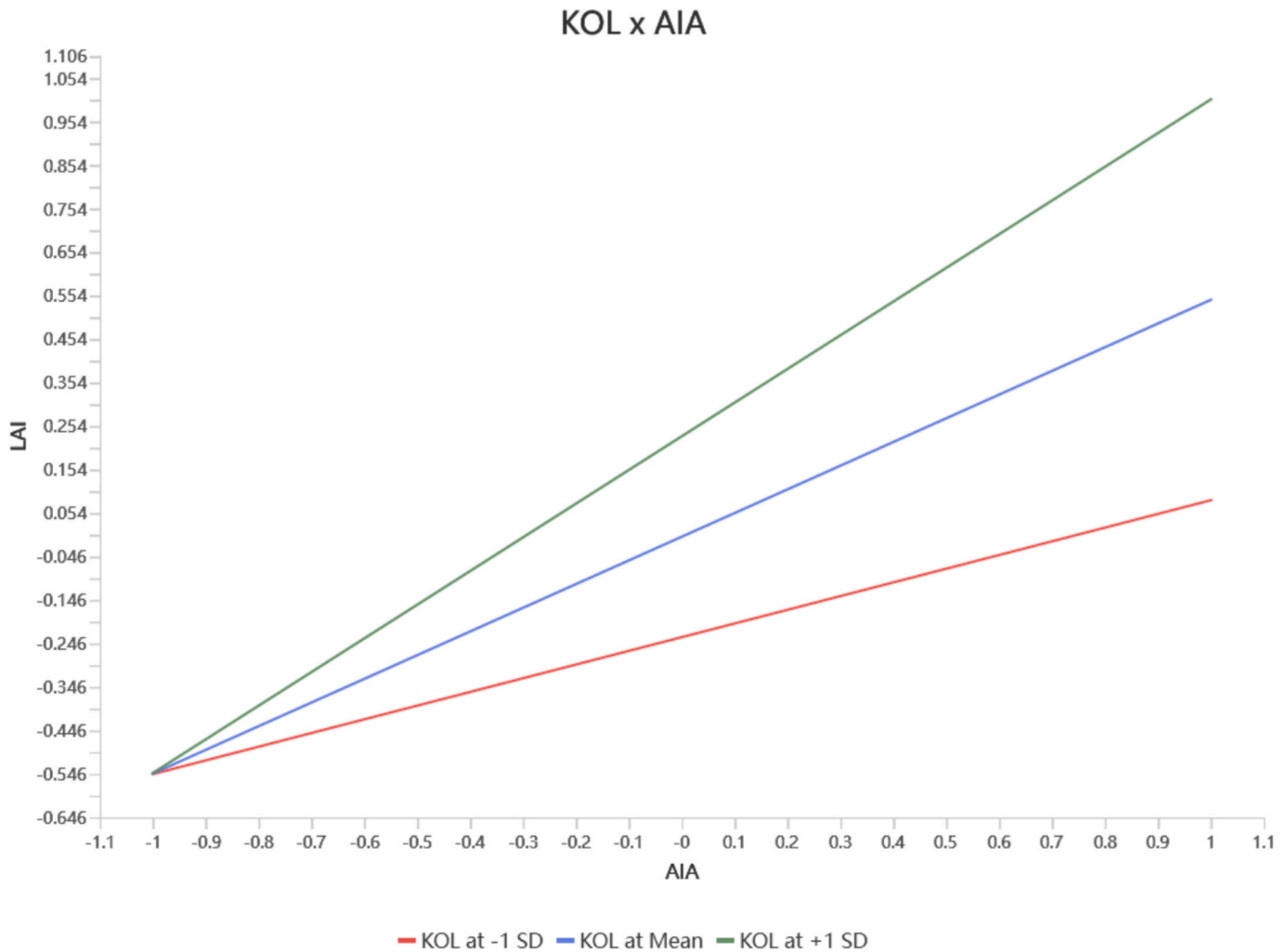


Fig. 4. Simple slope for moderating effect of KOL on AIA and LAI relation.

Table 8

Effect size (f^2).

Predictor constructs/dependent constructs	PSE	LAI	AP	MAIP
AIA	0.333 ^a	0.337 ^a	0.281 ^a	0.076 ^b
PSE			0.145 ^b	
LAI				0.052 ^b

Note: $0.15 < a < 0.35$, medium; $0.02 < b < 0.15$, small.

user's communication context, or is entirely fabricated yet appears reasonable but actually contradicts facts based on training data, users may experience artificial intelligence hallucinations (Onder and McCabe, 2025). Artificial intelligence hallucinations arise from users' trust in artificial intelligence, combined with artificial intelligence's "concealment and disguise", leading to misperceptions about artificial intelligence-generated content (such as knowledge). Currently, various artificial intelligence models such as ChatGPT and DeepSeek all exhibit artificial intelligence hallucinations to varying degrees. For employees with poor identification abilities or significantly lower efficiency than artificial intelligence, who thus highly trust and rely on artificial intelligence, even if they suspect the content generated by artificial intelligence, they may still fall into artificial intelligence hallucinations due to a lack of verification capabilities or resources (Sood et al., 2025). Therefore, in artificial intelligence adoption at work scenarios, there may be some employees who are unable to effectively cope with artificial intelligence and use it problematically (Hu et al., 2023) or whose

Table 9

Control variables.

Variables	Adaptive performance			Mimetic artificial intelligence performance		
	β	t-values	p-values	β	t-values	p-values
Gender	0.16	1.761	0.039	-0.025	0.221	0.412
Age	-0.025	0.536	0.296	0.089	1.738	0.041
Working seniority	0.067	1.495	0.067	0.031	0.667	0.252
Prior using artificial intelligence experience	0.033	0.738	0.23	0.058	1.067	0.143
Role	0.03	0.839	0.201	0.029	0.623	0.267
Position	0.064	1.305	0.096	0.045	0.87	0.192
Organizational artificial intelligence maturity	-0.014	0.366	0.357	-0.064	1.33	0.092
Individual technology readiness	0.019	0.446	0.328	-0.031	0.694	0.244

work behavior is completely dominated by artificial intelligence. For these employees, the conclusions discussed in this study may not apply, as social comparison between employees and artificial intelligence is fundamentally a more proactive behavior adopted by employees to

Table 10
Model's explanatory power, predictive relevance and fit.

Variable	R ²	R ² adjusted	Q ²	RMSE	MAE	SRMR
PSE	0.368	0.362	0.318	0.834	0.590	0.042
LAI	0.333	0.326	0.293	0.847	0.643	
AP	0.419	0.399	0.328	0.825	0.572	
MAIP	0.218	0.192	0.165	0.922	0.664	

Table 11
CVPAT results for out-of-sample predictive validity.

	Average loss difference	t-values	p-values
PLS-SEM vs indicator average (IA)			
Adaptive performance	−0.233	4.982	0
Employee learning from artificial intelligence	−0.203	5.589	0
Mimetic artificial intelligence performance	−0.108	3.065	0.002
Problem-solving efficacy	−0.202	4.689	0
Overall	−0.196	7.099	0
PLS-SEM vs linear model (LM)			
Adaptive performance	−0.092	2.923	0.004
Employee learning from artificial intelligence	−0.089	2.699	0.007
Mimetic artificial intelligence performance	−0.098	4.433	0
Problem-solving efficacy	−0.177	3.526	0
Overall	−0.114	6.113	0

adapt to artificial intelligence adoption at work.

After that clarification, we can now compare and discuss the conclusions of this study with existing research. Consistent with the broader literature, the results support Hypothesis 1, demonstrating that artificial intelligence adoption at work can significantly influence employee adaptive performance through problem-solving efficacy. This indicates that artificial intelligence technologies (such as generative artificial intelligence) can serve as effective tools for enhancing employees' confidence and efficiency in completing tasks within organizational settings. As Industry 5.0 and corporate digital transformation continue to deepen, the tool-like and practical nature of artificial intelligence is the primary reason for its adoption by businesses. While previous scholars have pointed out that artificial intelligence adoption may lead to negative impacts such as job insecurity and job displacement, this study proposes that after comparing themselves downward with artificial intelligence, employees will utilize artificial intelligence to enhance their problem-solving efficacy and adaptive performance. This finding confirms the

above hypothesis and also indicates that artificial intelligence adoption at work, even though its functional aspects may cause negative emotions among employees, should not be overlooked in terms of its potential positive impact on employee performance when actively regulated by employees. Additionally, this finding aligns with [Mo et al. \(2024\)](#), who emphasized the positive effect of problem-solving efficacy on enhancing individuals' problem-solving engagement and performance adaptability.

The results also suggest that learning from artificial intelligence, as a form of upward assimilation social comparisons between employees and artificial intelligence, impairs employees' mimetic artificial intelligence performance (H2). Mimetic artificial intelligence performance emphasizes employees leveraging the capabilities or skills they have learned from artificial intelligence to improve their work behavior and performance. Therefore, its origin lies in employees' learning behavior toward artificial intelligence. Unlike previous perspectives on artificial intelligence utilization, this research conclusion empirically demonstrates that as employees may engage in upward comparison with artificial intelligence and view it as a learning object. This not only demonstrates employees' self-regulatory capabilities when faced with artificial intelligence adoption at work but also highlights the diversity of roles artificial intelligence can assume in such contexts (e.g., artificial intelligence assistant, artificial intelligence colleague, artificial intelligence role model, etc.). Each role stems from employees' perceptions and attitudes toward artificial intelligence based on their specific circumstances, and these differing perceptions and attitudes in turn determine the varied impacts of artificial intelligence adoption at work on employees' work behavior and performance. Thus, this finding contributes to the understanding of this dynamic within artificial intelligence adoption at work environments, echoing [Li et al. \(2024\)](#), who discussed how employees' use of artificial intelligence triggers learning behaviors, while also responding to their calls for exploring the effects and consequences of employee learning from artificial intelligence. And this finding also fills a gap by providing empirical evidence for the mediating role of employee learning from artificial intelligence in the organizational context, an area that has been scarcely explored in existing literature.

Supporting Hypotheses 3 and 4, the findings highlight the moderating role of task-oriented leadership. Task-oriented leadership strengthens the positive effect of artificial intelligence adoption at work on both problem-solving efficacy and adaptive performance. This is consistent with the work of [Ahmed et al. \(2024\)](#), who identified the positive role of task-oriented leadership in facilitating employees' completion of tasks and enhancing performance outcomes. Similarly, this study confirms that the indirect effect of artificial intelligence adoption at work on mimetic artificial intelligence performance via

Table 12
RMSE values for ANN models.

	Model A		Model B		Model C		Model D	
	Input: AIA, PSE		Input: AIA, LAI		Input: AIA		Input: AIA	
	Output: AP		Output: MAIP		Output: PSE		Output: LAI	
	Training	Testing	Training	Testing	Training	Testing	Training	Testing
ANN network	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE
ANN1	0.0130	0.0089	0.0153	0.0197	0.0158	0.0170	0.0170	0.0124
ANN2	0.0126	0.0170	0.0156	0.0240	0.0169	0.0099	0.0170	0.0093
ANN3	0.0125	0.0128	0.0157	0.0147	0.0217	0.0204	0.0178	0.0302
ANN4	0.0128	0.0137	0.0157	0.0131	0.0158	0.0168	0.0172	0.0196
ANN5	0.0133	0.0104	0.0159	0.0183	0.0166	0.0227	0.0164	0.0165
ANN6	0.0131	0.0150	0.0153	0.0226	0.0153	0.0286	0.0165	0.0168
ANN7	0.0132	0.0087	0.0157	0.0120	0.0204	0.0120	0.0160	0.0170
ANN8	0.0128	0.0145	0.0163	0.0115	0.0169	0.0107	0.0160	0.0184
ANN9	0.0138	0.0150	0.0179	0.0096	0.0164	0.0131	0.0165	0.0145
ANN10	0.0121	0.0297	0.0191	0.0210	0.0164	0.0117	0.0166	0.0119
Mean	0.0129	0.0146	0.0163	0.0167	0.0172	0.0163	0.0167	0.0167
SD	0.0218	0.0774	0.0354	0.0715	0.0459	0.0779	0.0236	0.0757

Table 13
Sensitivity analysis.

Neural Network	Model A (Output: AP)		Model B (Output: MAIP)		Model C	Model D
	AIA	PSE	AIA	LAI	(Output: PSE)	(Output: LAI)
ANN1	0.611	0.389	0.517	0.483	1.000	1.000
ANN2	0.533	0.467	0.495	0.505	1.000	1.000
ANN3	0.543	0.457	0.531	0.469	1.000	1.000
ANN4	0.526	0.474	0.534	0.466	1.000	1.000
ANN5	0.505	0.495	0.499	0.501	1.000	1.000
ANN6	0.500	0.500	0.507	0.493	1.000	1.000
ANN7	0.505	0.495	0.520	0.480	1.000	1.000
ANN8	0.530	0.470	0.548	0.452	1.000	1.000
ANN9	0.496	0.504	0.520	0.480	1.000	1.000
ANN10	0.549	0.451	0.579	0.421	1.000	1.000
Average	0.530	0.470	0.525	0.475	1.000	1.000
relative importance						
Normalized	100.000	88.750	100.000	90.476	100.000	100.000
relative importance (%)						

Table 14
Comparison between PLS-SEM and ANN results.

PLS path	Path	Normalized	Ranking	Ranking	Remark
	Coefficient	relative importance (%)	(PLS-SEM)	(ANN)	
Model A (Output: AP)					
AIA → AP	0.438	100	1	1	Match
PSE → AP	0.320	88.750	2	2	Match
Model B (Output: MAIP)					
AIA → MAIP	0.280	100	1	1	Match
LAI → MAIP	0.233	90.476	2	2	Match
Model C (Output: PSE)					
AIA → PSE	0.564	100	1	1	Match
Model D (Output: LAI)					
AIA → LAI	0.545	100	1	1	Match

employee learning from artificial intelligence is conditionally influenced by knowledge-oriented leadership (H5, H6), reinforcing theories such as social learning theory and social comparison theory. This suggests that under the leadership of knowledge-oriented leaders, employees are more motivated and willing to imitate and learn from artificial intelligence, resulting in higher mimetic artificial intelligence performance. This finding resonates with [González-Mohino et al. \(2024\)](#) and [Dorta-Afonso et al. \(2024\)](#), emphasizing the positive influence of knowledge-oriented leadership on employee learning behaviors and their subsequent outcomes within organizations. In addition, the results of the control variable test indicate that employee age and gender are factors that influence work performance, which may be related to differences in cognitive processing when different employees encounter artificial intelligence technology ([Morris et al., 2005](#)).

5.1. Theoretical implications

This study makes a significant contribution to social comparison theory by empirically demonstrating how artificial intelligence adoption at work, as an internal digital transformation initiative, positively impacts employee performance (both adaptive performance and mimetic

artificial intelligence performance). By introducing problem-solving efficacy and learning from artificial intelligence as parallel mediators, this study enriches the understanding of the mechanisms through which artificial intelligence adoption at work affects employee performance and the process and outcomes of employee-artificial intelligence comparison within these mechanisms. It reveals that employees, when using artificial intelligence, enhance their confidence in utilizing artificial intelligence to solve problems and improve adaptive performance through downward comparison with artificial intelligence, while upward assimilation comparison with artificial intelligence increases their motivation to learn from artificial intelligence and boosts mimetic artificial intelligence performance. These findings provide a more nuanced interpretation of social comparison theory within the HAI context.

Furthermore, the study extends social learning theory by incorporating employee learning from artificial intelligence as a key factor mediating the relationship between artificial intelligence adoption at work and mimetic artificial intelligence performance. This addition underscores the importance of employee learning behaviors in the HAI context, suggesting that employees improve their performance by learning and imitating from artificial intelligence. This finding not only validates social learning theory within the organizational context but also expands its scope by emphasizing the role of learning from artificial intelligence, an aspect often underexplored within traditional social comparison theory. Additionally, this study verifies the mediating role of employee learning from artificial intelligence, highlighting that employees can achieve mimetic artificial intelligence performance through imitation and learning from artificial intelligence. By introducing this new variable, the research extends the existing body of work on employee performance in artificial intelligence application contexts.

Lastly, one of the unique theoretical contributions of this study is the combined use of PLS-SEM and ANN methods to explain the complex mechanisms through which artificial intelligence adoption at work affects employee performance. By integrating these two analytical methods, the research offers a comprehensive understanding of how artificial intelligence adoption at work influences employee performance through both linear and nonlinear pathways. This approach, employing PLS-SEM and ANN for data analysis, provides a more robust validation of the collective explanatory power of artificial intelligence adoption at work on employee performance, encouraging future research to adopt similarly integrative methods and perspectives.

5.2. Managerial implications

First, in manufacturing or operational contexts where repetitive tasks and process efficiency are central, framing artificial intelligence as a

collaborative learning partner can be achieved by providing role-specific artificial intelligence tools alongside hands-on simulations. For example, frontline workers in production lines could use artificial intelligence-driven digital twins to simulate equipment troubleshooting, with generative artificial intelligence providing real-time guidance on problem-solving steps. This aligns with Industry 5.0's emphasis on smart manufacturing, turning artificial intelligence into a "skill co-developer" that reduces job insecurity by linking artificial intelligence use directly to mastery of critical operational skills (e.g., predictive maintenance). Managers can further reinforce this by tying artificial intelligence-aided problem-solving to performance metrics like downtime reduction, making adaptive performance tangible in operational terms.

Second, in knowledge-intensive sectors (e.g., finance, consulting), a dual leadership approach should be tailored to the sector's focus on innovation and client-driven outcomes. Task-oriented leaders might set clear benchmarks for artificial intelligence-augmented tasks while knowledge-oriented leaders could host cross-team "artificial intelligence success clinics," where analysts share how they adapted artificial intelligence's data-sorting logic to refine client reports. This not only positions artificial intelligence as a role model for precision (mirroring consultants' need for accuracy) but also encourages mimetic artificial intelligence performance, such as adopting artificial intelligence's structured reasoning to present insights more persuasively to clients.

Third, in customer-centric industries (e.g., retail, healthcare), where demographic diversity among both employees and end-users is pronounced, inclusive artificial intelligence strategies must account for varied employee interactions with technology and client needs. For instance, in healthcare, older nurses might benefit from artificial intelligence tools with simplified interfaces paired with peer mentorship from tech-savvy colleagues, reducing resistance and building confidence in using artificial intelligence to personalize patient care. Conversely, younger retail staff could lead pilot programs for artificial intelligence chatbots, leveraging their familiarity with digital tools to refine the bot's customer service logic—turning age-related differences into a strength by fostering cross-generational knowledge exchange. This ensures artificial intelligence adoption enhances, rather than hinders, employees' ability to adapt to diverse customer demands.

5.3. Limitations and directions for future research

This research, while providing insightful findings, is not without its limitations.

First, one limitation of this study is that it primarily focuses on the relationship between artificial intelligence adoption at work and employee performance in specific internet companies context, which may limit the generalizability of the findings to other industries or work environments. Future research could explore how these results apply in different sectors or cultural contexts to see if similar patterns hold.

Second, the collection of data in Chinese scenarios may indeed introduce biases and affect the generalizability of the results. Chinese culture is the primary influencing factor. For example, the "face" (*mianzi*) culture may prompt participants to answer questionnaires inconsistently with their actual circumstances. This is driven by the desire to cater to the expectations of superiors or colleagues, demonstrate their own progress despite the implemented anonymity, or maintain a positive image of the organization when applying advanced artificial intelligence technologies. And it can lead to biases in the

questionnaire results. Additionally, the three-wave research design adopted in this study carries the risk that employees might communicate with each other during the intervals between waves. This communication could be influenced by the characteristics of collectivism and harmony in Chinese culture, which foster a strong willingness among employee teams to reach consensus, thereby compromising the accuracy of subsequent variable measurements. Future researchers can mitigate these biases by combining objective data (instead of relying entirely on self-reported data) and incorporating cross-cultural samples during the research phase.

Third, the research results indicate that problem-solving efficacy and employee learning from artificial intelligence both exert partial mediating effects, suggesting that there may be other potential mediators, such as emotional responses to artificial intelligence or job-specific skill development, which could further explain the artificial intelligence-performance relationship. Future research can build on this foundation by incorporating other variables for further in-depth study.

Fourth, while the consistency of the PLS-SEM and ANN analysis results in this study demonstrates the predictive power of PLS-SEM and the robustness of the test results, it also makes it hard to determine exactly what non-linear relationship the ANN captured. Future research can further investigate the limitations of the PLS-SEM and ANN data analysis methods in practical applications in light of the above circumstances.

6. Conclusion

This study, integrating social comparison, social learning and social influence theories, employs a mixed-method approach combining PLS-SEM and ANN, leading to the following conclusions: a) Artificial intelligence adoption at work is positively associated with employees' adaptive performance through problem-solving efficacy; b) Stronger artificial intelligence adoption at work is associated with an increased employees' mimetic artificial intelligence performance through employee learning from artificial intelligence; c) Task-oriented leadership strengthens both the positive effect of artificial intelligence adoption at work on employees' problem-solving efficacy and its mediating role. Similarly, knowledge-oriented leadership amplifies both the positive effect of artificial intelligence adoption at work on employees' learning from artificial intelligence and the mediating effect of this learning.

CRedit authorship contribution statement

Shengmin Liu: Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Yu Mei:** Writing – original draft, Methodology, Investigation, Data curation.

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Declaration of competing interest

The authors have no relevant financial or non-financial interests to disclose.

Appendix A. Simple comparison of the path analysis results between CB-SEM and PLS-SEM ($N = 308$)

Paths	CB-SEM			PLS-SEM		
	β	p-Values	95 % CI	β	p-Values	95 % CI
AIA - > PSE	0.459	0	[0.290, 0.617]	0.564	0	[0.462, 0.669]
AIA - > LAI	0.613	0	[0.477, 0.734]	0.545	0	[0.456, 0.626]
AIA - > AP	0.540	0	[0.347, 0.714]	0.438	0	[0.297, 0.560]
AIA - > MAIP	0.403	0.007	[0.157, 0.676]	0.280	0.001	[0.127, 0.421]
PSE - > AP	0.295	0.014	[0.108, 0.477]	0.320	0	[0.193, 0.451]
LAI - > MAIP	0.211	0.205	[-0.075, 0.450]	0.233	0.004	[0.081, 0.368]
AIA - > PSE - > AP	0.127	0.007	[0.059, 0.250]	0.180	0	[0.113, 0.265]
AIA - > LAI - > MAIP	0.115	0.170	[-0.031, 0.256]	0.127	0.003	[0.048, 0.203]

Note: The CB-SEM output was generated using AMOS 26.0, and the PLS-SEM output was generated using SmartPLS 4. The 95 % confidence intervals are all the Bias-corrected confidence intervals. AIA: Artificial intelligence adoption , PSE: Problem-solving efficacy, AP: Adaptive performance, LAI: Employee learning from artificial intelligence, MAIP: Mimetic artificial intelligence performance.

Data availability

Data will be available on request.

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